Econometrics: Methods and Applications
Test Exercise 6

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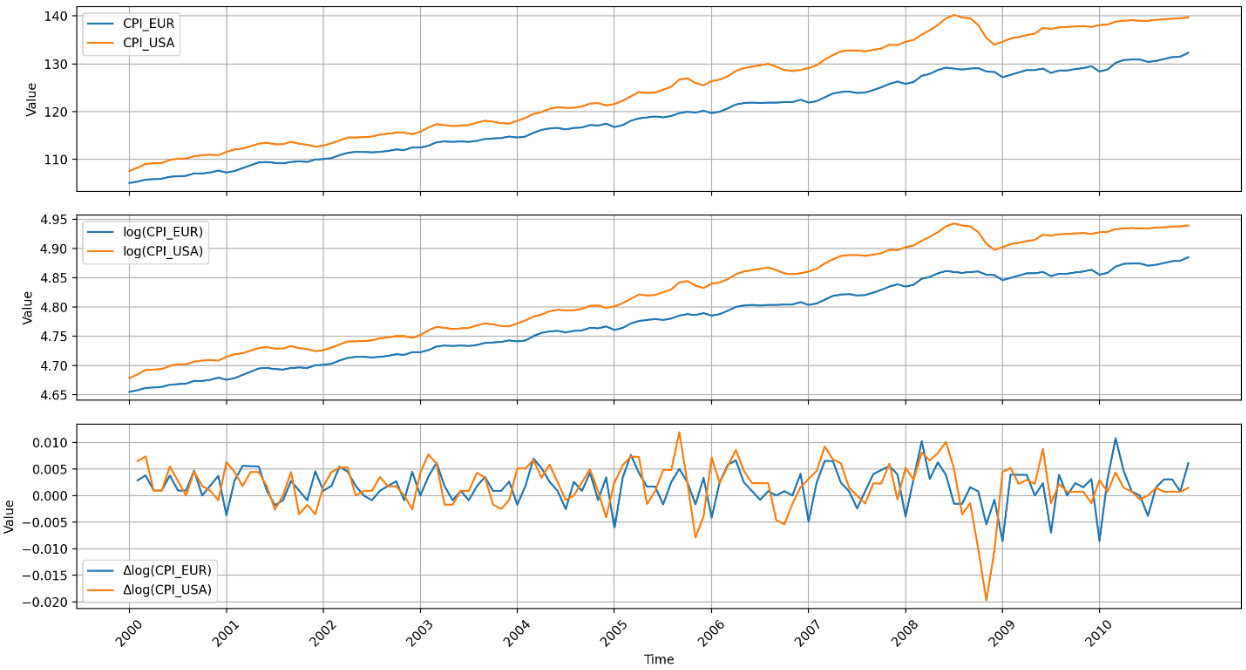
Note: This PDF presents the final numerical results and interpretation for Test Exercise 4. All regressions, statistical tests and numerical computations were implemented in Python. The corresponding source code for this exercise is available in Module_7/scripts/test_ex_6.py, which is included in the GitHub repository referenced above. The code is extensively documented with comments explaining the methodology, implementation, and interpretation of the results. The repository also contains the training exercises and supporting work completed throughout this module.

a) Below you can find the time series plots of:

- CPI_EUR and CPI_USA
- log(CPI_EUR) and log(CPI_USA)
- Δlog(CPI_EUR) and Δlog(CPI_USA)



Conclusion: CPI and log(CPI) series of both regions are trending upward and are therefore likely non-stationary, while the monthly inflation series fluctuate around a roughly constant mean. The inflation series DP show some common movements, especially during volatile periods such as 2009, but a clear and systematic visual lead-lag pattern from USA inflation to Euro inflation is not obvious. Therefore, whether USA inflation has predictive power for Euro-area inflation should be investigated formally.

b)

The ADF-test for log(CPI_EUR) yields:

- $\hat{\rho} = -0.12$
- $SE(\hat{\rho}) = 0.049$
- $t_{\hat{\rho}} = -2.455$

Therefore, since $t_{\hat{\rho}} = -2.455 > -3.5$, which is the 5% critical value when a deterministic trend is included, we fail to reject the null of non-stationarity for log(CPI_EUR).

The ADF-test for log(CPI_USA) yields:

- $\hat{\rho} = -0.071$
- $SE(\hat{\rho}) = 0.03$
- $t_{\hat{\rho}} = -2.403$

Therefore, since $t_{\hat{\rho}} = -2.403 > -3.5$, we also fail to reject the null of non-stationarity for log(CPI_USA).

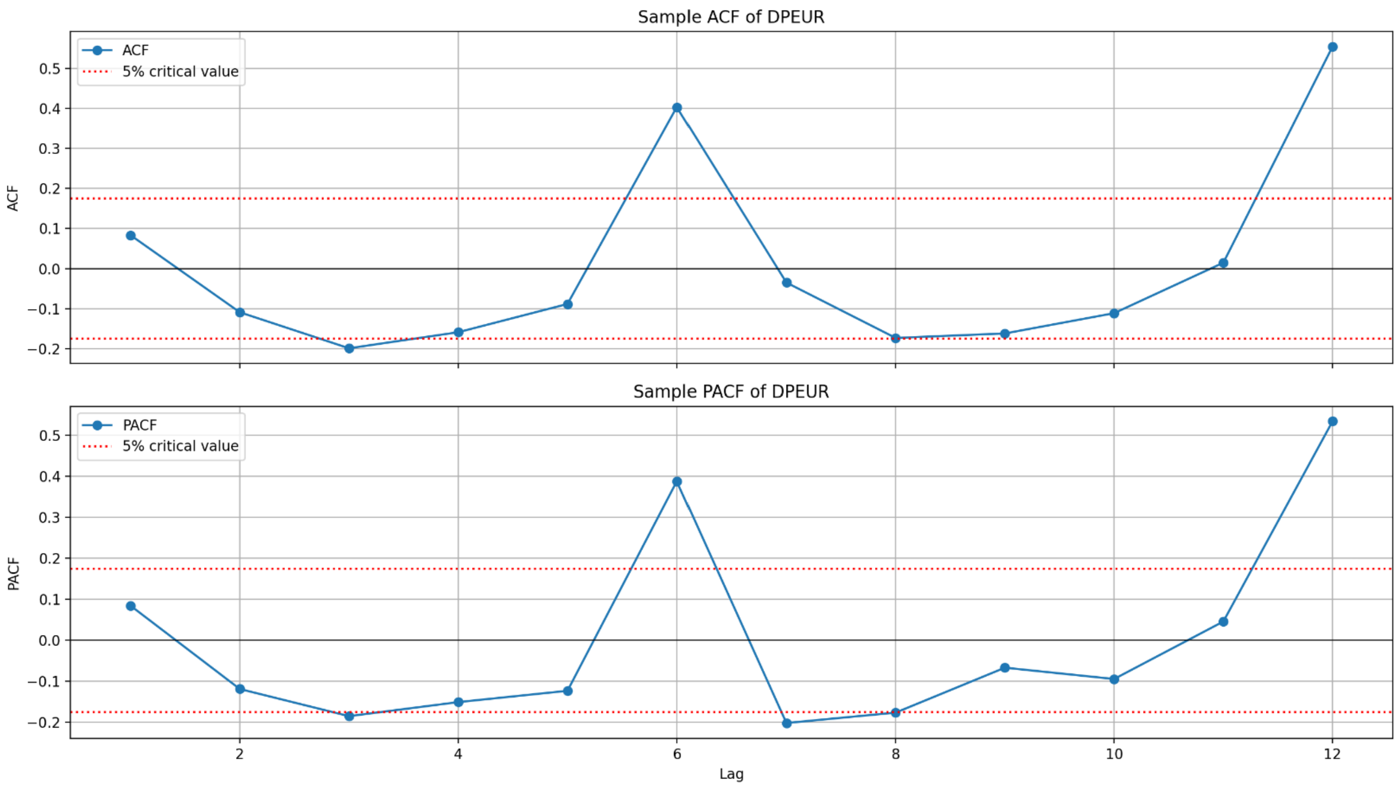
→ Both log series are non-stationary

c)

Using Python, we estimate the sample autocorrelations and partial autocorrelations of the series $\Delta \log(CPI\_EUR)$.

We use 12 lags for both of these calculations.

The 5% critical value is $\pm \frac{2}{\sqrt{131}} = \pm 0.175$



The ACF and PACF plots show pronounced and statistically significant spikes at lags 6 and 12. Since the data are monthly, lag 6 corresponds to a half-year seasonal effect, and lag 12 to an annual seasonal effect. This provides motivation for use of the following AR model:

$$DPEUR_t = \alpha + \beta_1 DPEUR_{t-6} + \beta_2 DPEUR_{t-12} + \varepsilon_t$$

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The AR model with lags 6 and 12 for DPEUR yields:

constant ->     β = 0.0         SE = 0.0        t-value = 1.365         p-value = 0.175

Lag6 ->         β = 0.189       SE = 0.077      t-value = 2.442         p-value = 0.016

Lag12 ->        β = 0.598       SE = 0.084      t-value = 7.157         p-value = 0.0

Therefore, both lag 6 and lag 12 are statistically significant at the 5% level.

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d) Extension of AR model to ADL model by adding lagged values of monthly inflation in the USA:

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The ADL model with lags 6 and 12 for DPEUR and lags 1, 6, and 12 for DPUSA yields:

constant ->     β = 0.0         SE = 0.0        t-value = 1.007         p-value = 0.316

EUR_Lag6 ->     β = 0.208       SE = 0.081      t-value = 2.584         p-value = 0.011

EUR_Lag12 ->    β = 0.673       SE = 0.089      t-value = 7.587         p-value = 0.0

USA_Lag1 ->     β = 0.231       SE = 0.051      t-value = 4.495         p-value = 0.0

USA_Lag6 ->     β = -0.042      SE = 0.055      t-value = -0.76         p-value = 0.449

USA_Lag12 ->    β = -0.248      SE = 0.054      t-value = -4.559        p-value = 0.0

Therefore, the coefficient of DPUSA at lag 6 is not statistically significant at the 5% level, we remove it from the model.
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The ADL model with lags 6 and 12 for DPEUR and lags 1 and 12 for DPUSA yields:

constant ->     β = 0.0         SE = 0.0        t-value = 1.267         p-value = 0.208

EUR_Lag6 ->     β = 0.169       SE = 0.071      t-value = 2.374         p-value = 0.019

EUR_Lag12 ->    β = 0.655       SE = 0.086      t-value = 7.651         p-value = 0.0

USA_Lag1 ->     β = 0.233       SE = 0.051      t-value = 4.582         p-value = 0.0

USA_Lag12 ->    β = -0.226      SE = 0.054      t-value = -4.189        p-value = 0.0

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e) Using the models of parts (c) and (d), we predict monthly Euro inflation for 2011, and then compute the values of RMSE, MAE, and SUM for each of the two forecast series.

The monthly inflation forecasts for 2011 using the AR model are:

YYYY-MM	DPEUR_AR
2011M01	-0.005
2011M02	0.003
2011M03	0.007
2011M04	0.004
2011M05	0.001
2011M06	0.002
2011M07	-0.003
2011M08	0.002
2011M09	0.005
2011M10	0.003
2011M11	0.001
2011M12	0.004

The monthly inflation forecasts for 2011 using the ADL model are:

YYYY-MM	DPEUR_ADL
2011M01	-0.006
2011M02	0.004
2011M03	0.008
2011M04	0.006
2011M05	0.002
2011M06	0.003
2011M07	-0.004
2011M08	0.002
2011M09	0.005
2011M10	0.004
2011M11	0.000
2011M12	0.004

Model evaluation table:

	AR	ADL
RMSE (x100) ->	0.232	0.211
MAE (x100) ->	0.169	0.14
SUM (x100) ->	0.506	0.048

The forecast evaluation results indicate that the ADL model performs better than the pure AR model for forecasting EUR inflation in 2011. All of the error metrics have lower values for the ADL model. These results suggest that, adding lagged USA inflation improves the out-of-sample performance of forecasting the EUR inflation. Therefore, it seems that USA inflation has predictive power for EUR inflation, at least in the 2011 forecast evaluation sample.